

8-(MathAA/10_HL_Winter_2021_Q3)-Exponentials-Logarithms

[Maximum mark: 5]

Solve the equation $\log_3 \sqrt{x} = \frac{1}{2\log_2 3} + \log_3(4x^3)$, where $x > 0$.



1-(MAT/21_HL_Summer_2016_Q5)-Graphs

The function f is given by $f(x) = \frac{3x^2+10}{x^2-4}$, $x \in \mathbb{R}$, $x \neq 2$, $x \neq -2$.

- (a) Prove that f is an even function. [2]
- (b) (i) Sketch the graph $y = f(x)$.
- (ii) Write down the range of f . [5]

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2-(MAT/21_HL_Summer_2017_Q6)-Graphs

Consider the graph of $y = \tan\left(x + \frac{\pi}{4}\right)$, where $-\pi \leq x \leq 2\pi$.

- (a) Write down the equations of the vertical asymptotes of the graph. [2]

The graph is reflected in the y -axis, then stretched parallel to the y -axis by a factor $\frac{1}{2}$,

then translated by $\begin{pmatrix} \frac{\pi}{4} \\ -3 \end{pmatrix}$.

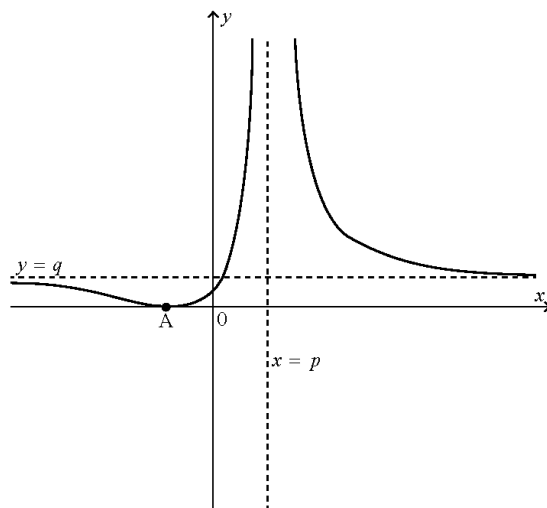
- (b) Give the equation of the transformed graph. [4]



3-(MAT/20_HL_Winter_2018_Q8)-Graphs

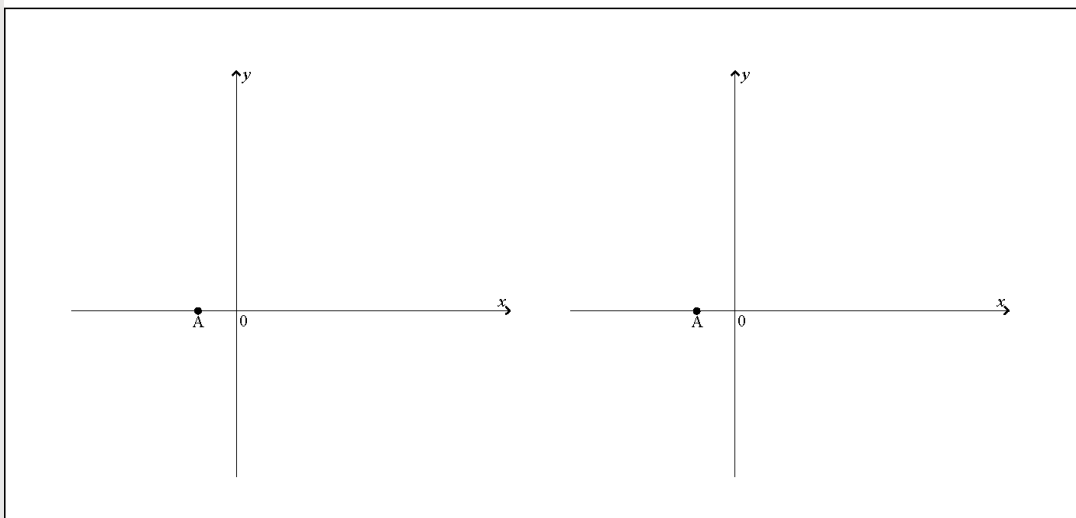
Consider the function $f(x) = \frac{ax+1}{bx+c}$, $x \neq -\frac{c}{b}$, where $a, b, c \in \mathbb{Z}$.

The following graph shows the curve $y = (f(x))^2$. It has asymptotes at $x = p$ and $y = q$ and meets the x -axis at A.



- (a) On the following axes, sketch the two possible graphs of $y = f(x)$ giving the equations of any asymptotes in terms of p and q .

[4]



- (b) Given that $p = \frac{4}{3}$, $q = \frac{4}{9}$ and A has coordinates $\left(-\frac{1}{2}, 0\right)$, determine the possible sets of values for a, b and c .

[4]



4-(MAT/22_HL_Summer_2019_Q9)-Graphs

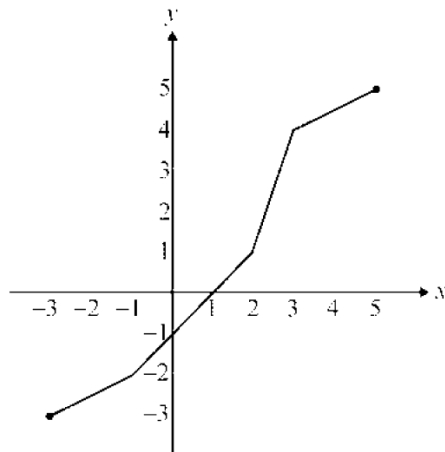
Consider the polynomial $P(z) \equiv z^4 - 6z^3 - 2z^2 + 58z - 51$, $z \in \mathbb{C}$.

- (a) Express $P(z)$ in the form $(z^2 + az + b)(z^2 + cz + d)$ where $a, b, c, d \in \mathbb{R}$. [7]
- (b) Sketch the graph of $y = x^4 - 6x^3 - 2x^2 + 58x - 51$, stating clearly the coordinates of any maximum and minimum points and intersections with axes. [6]
- (c) Hence, or otherwise, state the condition on $k \in \mathbb{R}$ such that all roots of the equation $P(z) = k$ are real. [2]



5-(MAT/20_HL_Winter_2019_Q3)-Graphs

[Maximum mark: 6]

The following diagram shows the graph of $y = f(x)$, $-3 \leq x \leq 5$.

- (a) Find the value of $(f \circ f)(1)$. [2]
- (b) Given that $f^{-1}(a) = 3$, determine the value of a . [2]
- (c) Given that $g(x) = 2f(x - 1)$, find the domain and range of g . [2]

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6-(MAT/10_HL_Winter_2020_Q10)-Differentiation, Graphs

Consider the function $f(x) = ax^3 + bx^2 + cx + d$, where $x \in \mathbb{R}$ and $a, b, c, d \in \mathbb{R}$.

- (a) (i) Write down an expression for $f'(x)$.
(ii) Hence, given that f^{-1} does not exist, show that $b^2 - 3ac > 0$. [4]

(b) Consider the function $g(x) = \frac{1}{2}x^3 - 3x^2 + 6x - 8$, where $x \in \mathbb{R}$.

- (i) Show that g^{-1} exists.
(ii) $g(x)$ can be written in the form $p(x - 2)^3 + q$, where $p, q \in \mathbb{R}$.
Find the value of p and the value of q .
(iii) Hence find $g^{-1}(x)$. [8]

The graph of $y = g(x)$ may be obtained by transforming the graph of $y = x^3$ using a sequence of three transformations.

- (c) State each of the transformations in the order in which they are applied. [3]
(d) Sketch the graphs of $y = g(x)$ and $y = g^{-1}(x)$ on the same set of axes, indicating the points where each graph crosses the coordinate axes. [5]



7-(MAT/10_HL_Winter_2020_Q12)-Graphs, Integration

Consider the function defined by $f(x) = \frac{kx-5}{x-k}$, where $x \in \mathbb{R} \setminus \{k\}$ and $k \neq 5$.

- (a) State the equation of the vertical asymptote on the graph of $y = f(x)$. [1]
- (b) State the equation of the horizontal asymptote on the graph of $y = f(x)$. [1]
- (c) Use an algebraic method to determine whether f is a self-inverse function. [4]

Consider the case where $k = 3$.

- (d) Sketch the graph of $y = f(x)$, stating clearly the equations of any asymptotes and the coordinates of any points of intersections with the coordinate axes. [3]
- (e) The region bounded by the x -axis, the curve $y = f(x)$, and the lines $x = 5$ and $x = 7$ is rotated through 2π about the x -axis. Find the volume of the solid generated, giving your answer in the form $\pi(a + b \ln 2)$, where $a, b \in \mathbb{Z}$. [6]



8-(MathAA/10_HL_Winter_2021_Q2)-Algebra-functions-Roots, Graphs

[Maximum mark: 9]

The function f is defined by $f(x) = \frac{2x+4}{3-x}$, where $x \in \mathbb{R}$, $x \neq 3$.

(a) Write down the equation of

- (i) the vertical asymptote of the graph of f ;
- (ii) the horizontal asymptote of the graph of f .

[2]

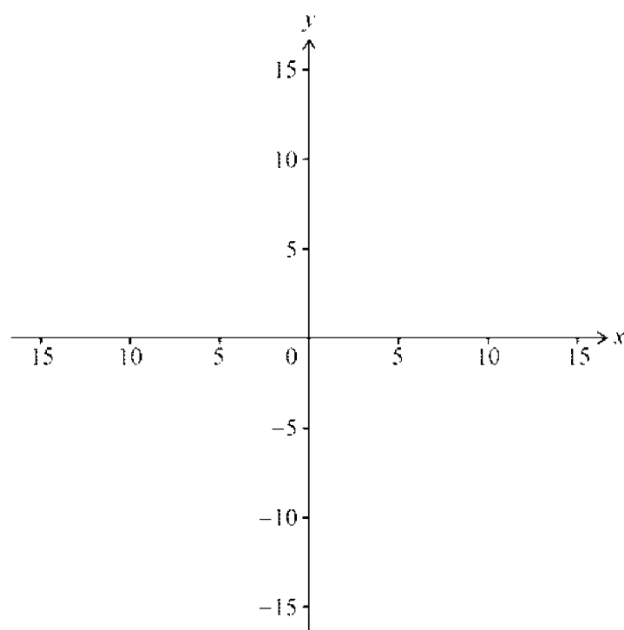
(b) Find the coordinates where the graph of f crosses

- (i) the x -axis;
- (ii) the y -axis.

[2]

(c) Sketch the graph of f on the axes below.

[1]



The function g is defined by $g(x) = \frac{ax+4}{3-x}$, where $x \in \mathbb{R}$, $x \neq 3$ and $a \in \mathbb{R}$.

(d) Given that $g(x) = g^{-1}(x)$, determine the value of a .

[4]



9-(MathAA/20_HL_Winter_2021_Q10)-Algebra-functions-Roots, Graphs, Integration

[Maximum mark: 18]

Consider the function $f(x) = \frac{x^2 - x - 12}{2x - 15}$, $x \in \mathbb{R}$, $x \neq \frac{15}{2}$.

(a) Find the coordinates where the graph of f crosses the

(i) x -axis;

(ii) y -axis.

[3]

(b) Write down the equation of the vertical asymptote of the graph of f .

[1]

(c) The oblique asymptote of the graph of f can be written as $y = ax + b$ where $a, b \in \mathbb{Q}$.

Find the value of a and the value of b .

[4]

(d) Sketch the graph of f for $-30 \leq x \leq 30$, clearly indicating the points of intersection with each axis and any asymptotes.

[3]

(e) (i) Express $\frac{1}{f(x)}$ in partial fractions.

(ii) Hence find the exact value of $\int_0^3 \frac{1}{f(x)} dx$, expressing your answer as a single logarithm.

[7]



10–(MathAA/21_HL_Summer_2021_Q11)–Algebra-functions-Roots, Graphs

[Maximum mark: 20]

The function f is defined by $f(x) = \frac{3x+2}{4x^2-1}$, for $x \in \mathbb{R}$, $x \neq p$, $x \neq q$.

(a) Find the value of p and the value of q . [2]

(b) Find an expression for $f'(x)$. [3]

The graph of $y = f(x)$ has exactly one point of inflexion.

(c) Find the x -coordinate of the point of inflexion. [2]

(d) Sketch the graph of $y = f(x)$ for $-3 \leq x \leq 3$, showing the values of any axes intercepts, the coordinates of any local maxima and local minima, and giving the equations of any asymptotes. [5]

The function g is defined by $g(x) = \frac{4x^2-1}{3x+2}$, for $x \in \mathbb{R}$, $x \neq -\frac{2}{3}$.

(e) Find the equations of all the asymptotes on the graph of $y = g(x)$. [4]

(f) By considering the graph of $y = g(x) - f(x)$, or otherwise, solve $f(x) < g(x)$ for $x \in \mathbb{R}$. [4]



11–(MathAA/11_HL_Summer_2022_Q5)–Graphs

[Maximum mark: 7]

Consider $f(x) = 4 \sin x + 2.5$ and $g(x) = 4 \sin\left(x - \frac{3\pi}{2}\right) + 2.5 + q$, where $x \in \mathbb{R}$ and $q > 0$.

The graph of g is obtained by two transformations of the graph of f .

(a) Describe these two transformations. [2]

The y -intercept of the graph of g is at $(0, r)$.

(b) Given that $g(x) \geq 7$, find the smallest value of r . [5]



12-(MathAA/12_HL_Summer_2022_Q3)-Graphs

[Maximum mark: 8]

A function f is defined by $f(x) = \frac{2x-1}{x+1}$, where $x \in \mathbb{R}$, $x \neq -1$.

- (a) The graph of $y = f(x)$ has a vertical asymptote and a horizontal asymptote.

Write down the equation of

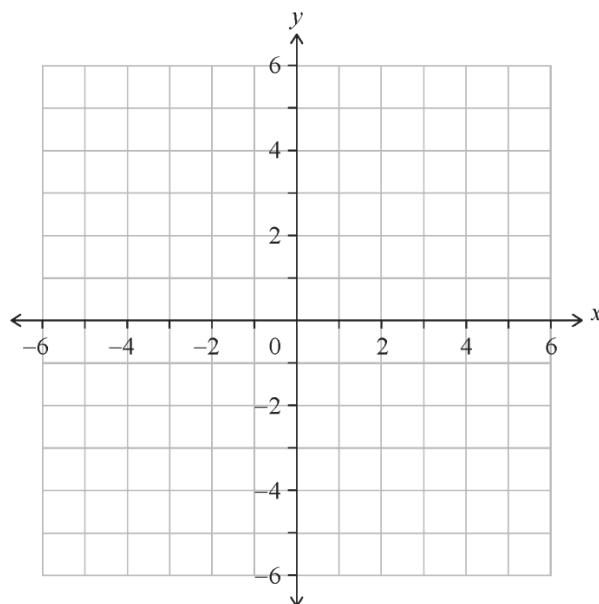
- (i) the vertical asymptote;
(ii) the horizontal asymptote.

[2]

- (b) On the set of axes below, sketch the graph of $y = f(x)$.

On your sketch, clearly indicate the asymptotes and the position of any points of intersection with the axes.

[3]



- (c) Hence, solve the inequality $0 < \frac{2x-1}{x+1} < 2$.

[1]

- (d) Solve the inequality $0 < \frac{2|x|-1}{|x|+1} < 2$.

[2]



13–(MathAA/10_HL_Winter_2022_Q10)–Differentiation, Graphs

[Maximum mark: 20]

The function f is defined by $f(x) = \cos^2 x - 3 \sin^2 x$, $0 \leq x \leq \pi$.

- (a) Find the roots of the equation $f(x) = 0$. [5]
- (b) (i) Find $f'(x)$. [4]
- (ii) Hence find the coordinates of the points on the graph of $y = f(x)$ where $f'(x) = 0$. [7]
- (c) Sketch the graph of $y = |f(x)|$, clearly showing the coordinates of any points where $f'(x) = 0$ and any points where the graph meets the coordinate axes. [4]
- (d) Hence or otherwise, solve the inequality $|f(x)| > 1$. [4]



1-(MAT/11_HL_Summer_2012_Q3)-ComplexNumbers

[Maximum mark: 7]

If $z_1 = a + a\sqrt{3}i$ and $z_2 = 1 - i$, where a is a real constant, express z_1 and z_2 in the form $r \operatorname{cis} \theta$, and hence find an expression for $\left(\frac{z_1}{z_2}\right)^6$ in terms of a and i .

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